

Interdependent conflict escalation and duration: Taking into account interdependencies between rebel organizations

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Overview

- ▶ Main argument: Need to take into account interdependencies between countries and organizations to:
 - ▶ explain spread of violence
 - ▶ explain interdependent dynamics (e.g. free-riding, outbidding etc)
 - ▶ and make accurate predictions
- ▶ Presentation includes work with
 - ▶ Julian Wucherpfennig (Hertie School of Governance)
 - ▶ Elio Amicarelli (UCL)
 - ▶ Altaf Ali (UCL)
 - ▶ Gokhan Ciflikli (LSE)
 - ▶ Martin Steinwand (Stony Brook (University of Essex))
 - ▶ Bahar Leventoglu (Duke University)
 - ▶ Michael D. Ward (Duke University)
 - ▶ Shahryar Minhas (Michigan State)
 - ▶ Cassy Dorff (University of New Mexico)

Actors and their Interactions

Theoretical advances and empirical limitations

Monadic analysis → Dyadic analysis → Network analysis



Collier and Hoeffler (2000)
Fearon and Laitin (2003)
Bates et al. (2005)
Cunningham (2006)
Gleditsch et al. (2002)
Hegre et al. (2001)

Cunningham et al. (2009)
Wucherpfennig et al. (2012)
Bapat and Bond (2012)
Walter (2009)
Bakke et al. (2012)
Steinwand (2011)
Cunningham (2013)

Current

How to think about interdependencies

Country or actor level

From a network perspective:

- ▶ Sender
- ▶ Receiver
- ▶ Dyad
- ▶ Second and third-order effects

How to think about interdependencies

Country or actor level

$$S \longrightarrow R$$

- ▶ Sender
- ▶ Receiver
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How to think about interdependencies

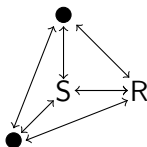
Country or actor level

$$S \longleftrightarrow R$$

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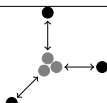
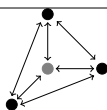
How to think about interdependencies

Country or actor level



- ▶ Sender
- ▶ Receiver
- ▶ Dyad
- ▶ Second and third-order effects

Table: Analyzing interdependencies and the link between theory (utility of units) and empirical model. Arrows describe strategic interdependencies between actors.

Type	Representation	Rebels' Utility	$y = \dots + \epsilon$
Aggregate		$U_i(X)$	$X\beta$
Dyadic		$U_i(X, Z)$	$X\beta + Z\zeta$
K-dyads		$U_i(X, Z, K)$	$X\beta + Z\zeta + K\kappa$
K-adic		$U_i(X, Z, Y_j, W)$	$X\beta + Z\zeta + K\kappa + \rho W\gamma$

Rebels and terrorist organizations are not alone

- ▶ Interdependencies in conflicts
 - ▶ Interdependent escalation
 - ▶ Interdependent duration

Multiple Actors and Civil War

Syrian Free Army

Coalition of:

- ▶ Syria Revolutionaries Front
- ▶ Hazzm Movement
- ▶ Southern Command
- ▶ First Army
- ▶ Yarmouk Brigade
- ▶ Dawn of Freedom Brigades

Multiple Actors and Civil War

Some stylized facts

- ▶ Civil conflicts typically involve several rebel groups and last longer (e.g. Cunningham, 2006).
- ▶ Rebel groups form coalitions to fight the government (e.g. Christia, 2012).
- ▶ Coalitions splinter, leading to continued fighting (e.g. Bakke et al, 2012).

Problems

- ▶ Policy related: Conflicts with more actors are harder to resolve (Cunningham 2006).
- ▶ Research literature: Existing theoretical & empirical models of civil war typically only look at **dyadic interactions**.

⇒ Need to account for strategic interactions to explain rebel fighting durations

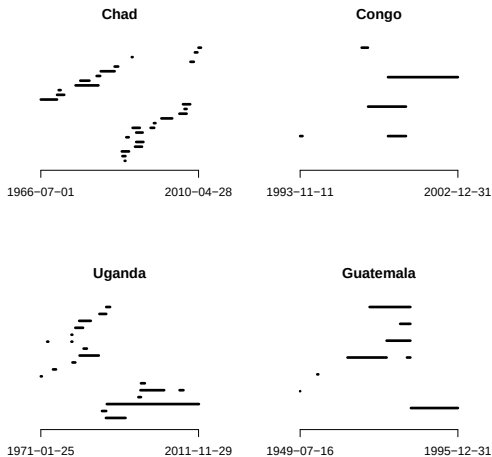
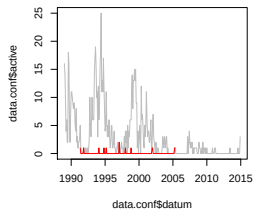
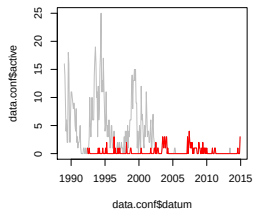
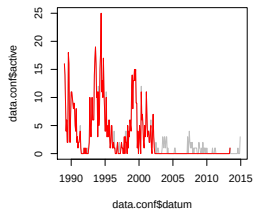
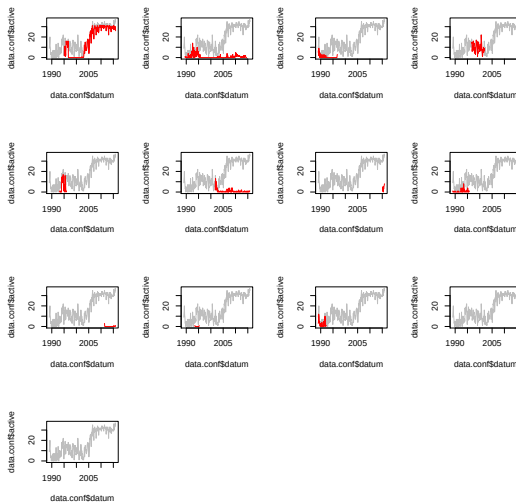


Figure: Rebel organization activity during selected UCDP armed conflicts. Each row represents a separate rebel organization as defined by a unique UCDP *SideB* for each rebel organization.

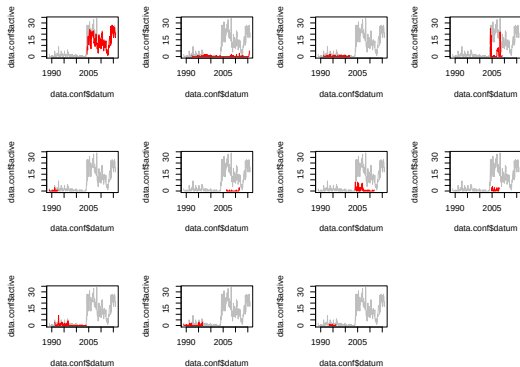
Interdependent escalation



Interdependent escalation



Interdependent escalation



Yet to Date...

Strategic interaction is not systematically integrated in formal and empirical models, despite widespread agreement that it matters (e.g. Walter 2009, Cunningham 2006).

Our Contribution (Metternich and Wucherpfennig, under review)

First study that provides *both* a

- ▶ theoretical model and
- ▶ an empirical model

of strategic rebel group interdependence.

- 1 Showing that civil wars are generally dominated by competition dynamics.
- 2 However, conflict of public good provision (e.g. separatist conflicts) we see less competition

Public Goods → Problems

- ▶ Fighting efforts against the government by one group that weaken the government will increase the chances of all other to obtain a favorable outcome
 - ▶ Rebel organizations formalize cooperation through alliances to avoid mutual exploitation (e.g., Bapat & Bond 2012)
- Rebel organizations face incentives to underprovide (free-ride).
- Fighting is a strategic substitute.

Why Compete?

- ▶ Cunningham, Bakke & Seymour 2012, Fjelde & Nillson 2012: political relevance, competition for resources (process-oriented)
 - ▶ Outcome-focused: → division of the spoils
 - ▶ Taking control of the post-conflict government: private or club good
 - ▶ Who gets what?
 - ① Contest over relative survival: “last rebel standing”
 - Outlast competitors
 - Signal strength (attrition)
 - ② Division of spoils related to relative contribution
 - Burden bearing compensated through spoils share
 - ▶ Competition through fighting the government
- Rebel organizations have incentives to over-provide

Spatial Econometrics for Durations

AFT Duration Model

$$y = \mathbf{X}\beta + \frac{1}{\lambda}u \quad (1)$$

where:

- ▶ y = logged time until failure
- ▶ X = covariates
- ▶ u = stochastic error (i.i.d), scaled by a shape parameter λ

Hays and Kachi, 2009

$$y = \rho \mathbf{W}y + \mathbf{X}\beta + \frac{1}{\lambda}u \quad (2)$$

- ▶ “Space is more than geography” (Beck et al., 2006)

UCDP/PRIO based dataset on rebel organizations (NSA) by Cunningham, Gleditsch and Salehyan (2009)

- ▶ 379 separate conflict dyads in 265 distinct conflicts
- ▶ 260 dyads in 58 conflicts with $n \geq 2$
- ▶ 107 dyads in secessionist conflicts
- ▶ 272 dyads in non-secessionist conflicts
- ▶ Entry and exit dates
- ▶ Various covariates

Model fit and insample prediction

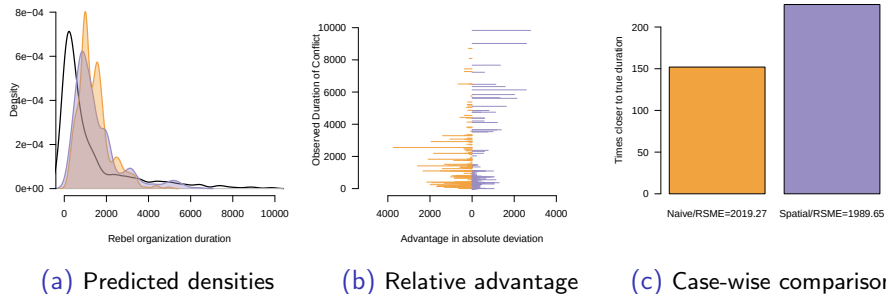
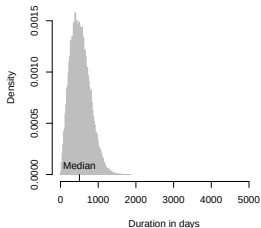
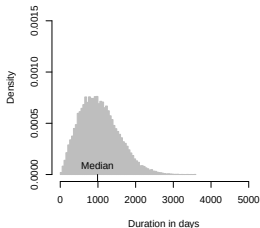


Figure: Model fit and insample prediction assessment comparing Model 1 (k-dyadic Weibull controlling for the number of rebel organizations) and Model 3 (spatial k-adic Weibull including the number of rebel organizations). In all panels orange refers to Model 1, while purple pertains to Model 3.

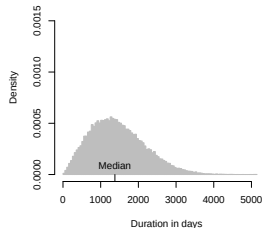
Figure: Simulated durations based on model estimates in a hypothetical conflict with four interdependent rebel organizations



(a) $\rho = 0$, Median= 502



(b) $\rho = 0.1$, Median= 990



(c) $\rho = 0.14$, Median= 1377

- ▶ Based on the insights of interdependent actors, we are developing two forecasting models
 - ▶ RebelCast Escalation
 - Country
 - Rebel
 - ▶ RebelCast Duration
 - Country
 - Rebel
- ▶ Both use data from RebelTrack
 - ▶ R package calculating monthly rebel organization measures based on UCDP GED
 - Area active
 - Event count
 - Distance to capital
 - Mean fighting position
 - Distance to other rebel organizations
 - etc

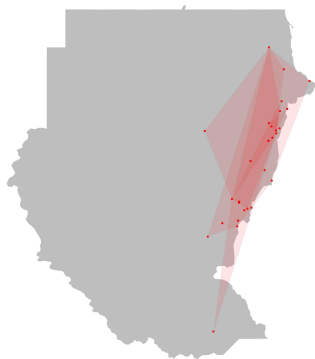


Figure: NDA. Hull area of fighting activity

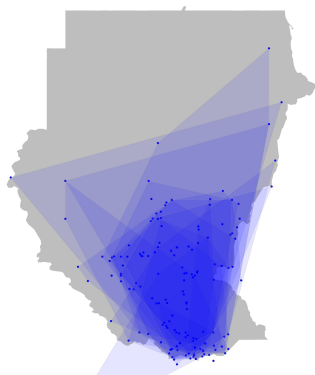


Figure: SPLM/A. Hull area of fighting activity

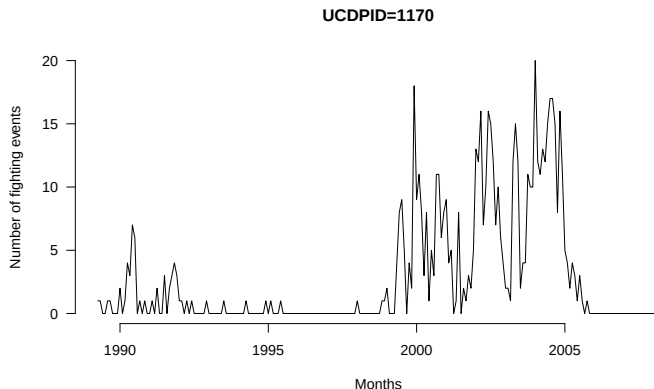
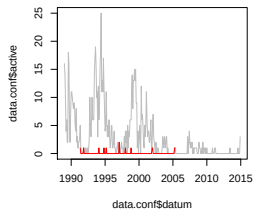
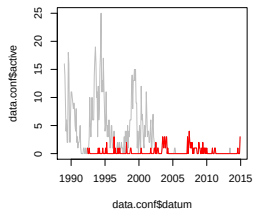
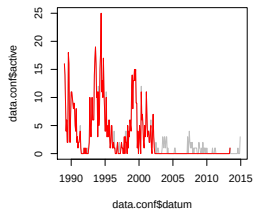


Figure: GAM. Number of active days per month

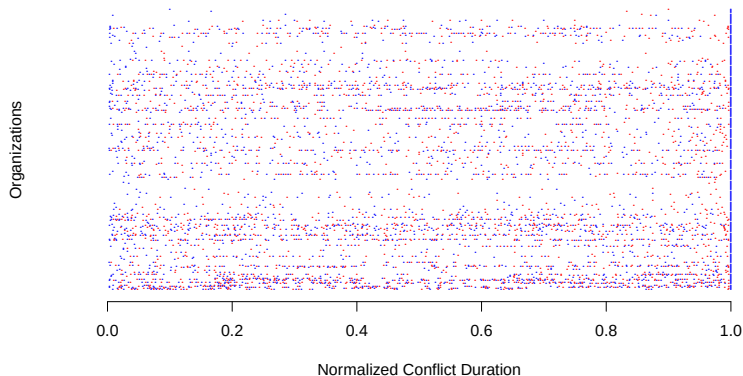
RebelCast Escalation



Machine learning approach

- ▶ Random Forests
- ▶ Outcome: UCDP-GED events (excluding one-side violence) per country-month
- ▶ Input:
 - ▶ RebelCast variables
 - ▶ ICEWS/Phoenix

Fighting and Interruption Spells



Machine learning approach

- ▶ Multi-state models
- ▶ Outcome: UCDP GED Rebel spells (End, Interruption, Fighting)
- ▶ Input
 - ▶ RebelCast variables
 - ▶ ICEWS/Phoenix

Conclusion and looking ahead

- ▶ Interdependencies matter (focused on rebels, but this extends to states, ethnic groups etc.)
- ▶ Need adequate data to capture organizational features and interdependencies
- ▶ Macro-level predictions are likely to be improved by understanding internal (e.g. rebels), transnational (e.g. trans-border cooperation), and international (e.g. interventions) dynamics